

Analytic UX Evaluation: Data Collection

Highlights

- Criticism of analytic methods
- Focus groups
- Design walk-throughs as early analytic evaluation methods
- Expert UX inspection
- Heuristic evaluation
- Our practical approach to UX inspection

23.1 INTRODUCTION

23.1.1 You Are Here

We begin each process chapter with a “You are here” picture of the chapter topic in the context of The Wheel, the overall UX design lifecycle template (Fig. 23.1). As an alternative approach to the empirical data collection methods of the previous chapter, this chapter is about analytic data collection methods.

Historically, analytic evaluation methods (e.g., UX inspection) have made their mark in small and domain-easy products and systems; however, as the UX discipline has evolved with more mature and tested interaction design patterns and standardized user interface toolkits and libraries, coupled with the demand for speed in agile processes, nimbler analytic methods have been edging out slower and bulkier empirical evaluation methods, even for large systems in complex domains. Experts now can examine these designs and be equally effective, at or least adequately effective, at identifying UX problems.

23.1.2 Adding Analytic Methods to the Mix

In Chapter 22, we saw that empirical evaluation methods were closely tied to users and usage experience. Analytic evaluation does not rely on users or usage experience. Analytic evaluation is based on examining inherent attributes of

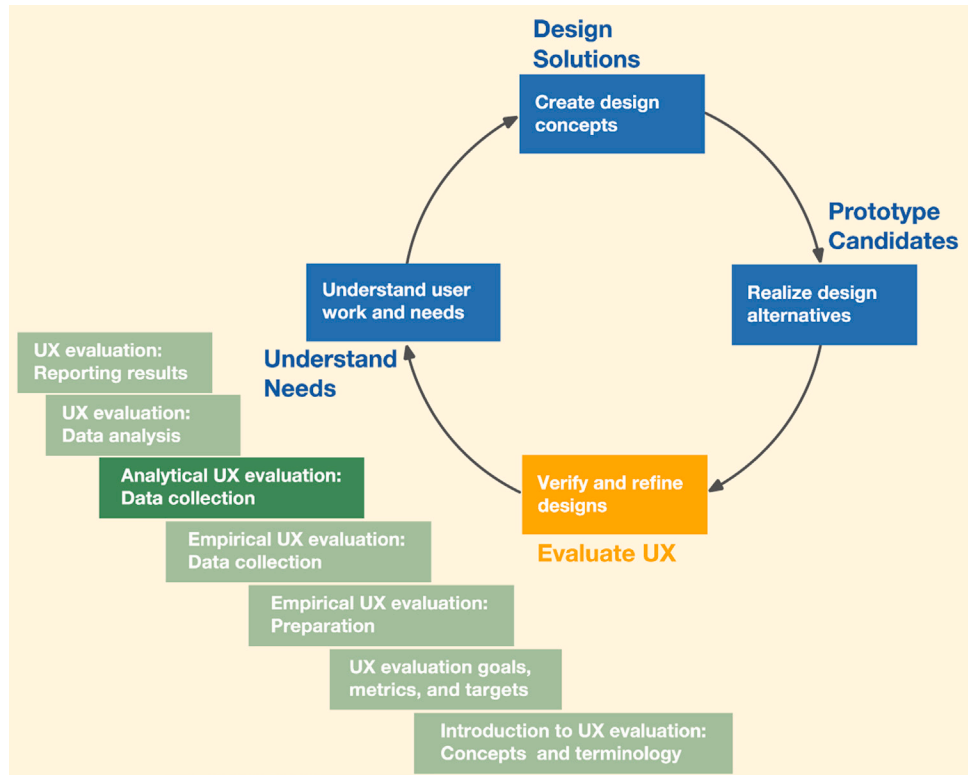


Fig. 23.1

You are here, in the chapter on analytic UX evaluation data collection within the evaluate UX activity, in the context of the overall Wheel UX lifecycle.

the design rather than seeing the design in use. The result is still formative data about UX problems to fix.

Although it is possible to perform analytic methods with high rigor and therefore less speed, they were developed as faster and less expensive methods to produce approximations to, or predictors of, empirical results. The reason they tend to be more rapid and less expensive is because you do not need to:

- Identify and recruit user participants
- Schedule participant sessions and bring in participants (or visit them)
- Run long task-based evaluation sessions

Following are general characteristics of analytic evaluation methods:

- Because they are based on expert opinions rather than empirical usage data, they require experts in both UX and the subject-matter domain (often different people working together).
- They are effective at finding important qualitative data, the UX problems that are cost-effective to fix right off the top.

- They are also good at finding deep-rooted conceptual problems, something empirical evaluation methods do not always do well. These are usually more difficult to deal with but are essential to fix.
- Most analytic UX evaluation methods can be used at any level of prototype fidelity. Experts are good at mentally filling in gaps in prototype fidelity.
- Usually analytic methods lean toward being less formal, with less protocol and fewer rules.
- There is much more variability in the process, with almost every evaluation session being different, tailored to the prevailing conditions.
- This freedom to adapt to conditions creates more room for spontaneous ingenuity, something experienced UX professionals do best.

23.1.3 Criticism of Analytic Methods: Discount Methods?

Analytic UX evaluation methods, including most inspection methods, have been disparaged in human–computer interaction (HCI) literature of the past. Sometimes disparagingly called “discount methods” (Gray & Salzman, 1998), analytic methods have been criticized for not being thorough and for being unscientific. Time has proved this criticism wrong, and now these methods, intended to be practical, not scientific, are the bread and butter of agile UX practice (see [Section 19.7.2.2](#)).

The rapid methods we discuss in this chapter are the result of trading off the thoroughness of high rigor for low cost, fast application, and effectiveness—a happy compromise for practical use in agile UX design environments.

23.1.4 Focus Groups

A focus group is a small discussion group of representative users or stakeholders aimed at identifying broad themes and issues in a work practice (see [Section 6.2.4](#)). Focus groups, composed of a moderator and, say, a half-dozen other participants, can be used as an analytic evaluation method for early evaluation of UX designs. As Martin and Hanington (2012) say, “The power of focus groups lies in the group dynamic that it creates.” Group members share opinions and collaboratively discuss the pros and cons as peers. As the group members describe their experiences and feelings, they will tell stories and use metaphors and analogies. At the end of a session, the moderator leads the group in creating a summary of the discussion.

Focus groups are mostly useful in assessing a group evaluation of high-level issues, such as brand or perceptions of one’s product compared to the competition. Because they can be expensive to run, require considerable

planning and recruiting, and often are not highly effective at getting detailed feedback, it is pretty rare these days to use focus groups in UX practice.

23.2 THE REALITY OF ANALYTIC UX EVALUATION METHODS

If, as you read this chapter, you are a student aspiring to be a UX practitioner or you are already a UX practitioner, then you should expect the terminology and practice of analytic evaluation methods to be varied and ad hoc, much of it invented to meet UX team evaluation needs of the moment and often given impromptu names or no names at all.

With that in mind, we present these analytic methods, taken from our own experience and from the HCI literature, as examples of what is possible and what can be adapted and appropriated as you synthesize your own methods to meet specific and immediate needs.

23.3 DESIGN WALK-THROUGHS

A design walk-through is an informal analytic formative evaluation method and is a kind of tour of the design features. Some UX teams call this a design review or a critique. In practice, sometimes a design walk-through is used more as communication than evaluation. For example, it is used to bring the software engineering team up to speed on a new UX design.

An activity providing a forum for the UX design team to present and demonstrate UX designs, a design walk-through is intended to obtain high-level design feedback from non-UX stakeholders, including customers, potential users, subject-matter experts, business representatives, and other broader cross-functional stakeholders as they participate in deep analysis and critiquing. The goal is to get initial reactions to design concepts and visions.

Although there are many other methods out there called walk-throughs, and they are used in various ways in UX practice, perhaps the best distinguisher of what we mean here is that a design walk-through is used to get the right UX design, whereas later evaluations are used to get the UX design right.

Design walk-throughs are especially useful for discussion of such designs as:

- Initial ideas on the ecology of the system
- Conceptual design

- Design ideas for engendering emotional impact
- Getting started with task-level designs
- Overall look and feel
- Visual, auditory, motion (animations, etc.) design ideas and concepts

23.3.1 Parameters for Design Walk-throughs

23.3.1.1 Goals for design walk-throughs

The goals at this stage are to be certain that the stakeholders involved understand and approve of the conceptual vision and high-level flow. You want to ensure that the emerging understanding of the design solution aligns with the business and user goals in the envisioned solution domain and that they satisfy any constraints from a technology perspective.

23.3.1.2 Expected feedback

As an evaluation method, design walk-throughs obtain feedback about the design, such as the answers to the following questions:

- Did this design walk-through help you understand the emerging design solution?
- Is the conceptual vision consistent with your expectation?
- Is the high-level flow demonstrated here appropriate?
- Does the early vision of the solution match your idea of what it should be?
- Does the emerging design solution align with the business and user goals in the envisioned solution domain?
- Does the design solution look like it will satisfy constraints from a technology perspective?
- Does the design have good support for emotional impact, including aesthetics and fun?

23.3.1.3 Artifacts involved in design walk-throughs

Artifacts most often used in design walk-throughs include early sketches, design scenarios, user stories, and storyboards (visual scenarios in the form of a series of sketches or graphical clips, often annotated, in cartoonlike frames, illustrating the interplay between a user and an envisioned ecology or device). Other artifacts in design walk-throughs include wireframes (and wireframe prototypes), object models, other design representations for task flow, and UX research data models.

23.3.2 Prepare for a Design Walk-through

Prepare for a design walk-through by doing the following:

- Know your design and avoid the distraction of getting confused about how it works.
- If useful, begin with storyboards to illustrate flow, and post these on the side.

- Have at hand descriptions of relevant users, work roles, and user classes; post these, too.
- Schedule the session with the appropriate stakeholders.
 - Your UX studio is a good venue for this.
- Decide who will take the role of note taker to record new ideas, barriers and challenges, and changes proposed, and ideas about the way forward.

23.3.3 Conduct a Design Walk-through Session

Informal session protocols can include the following:

- The presenter (one of the team, usually a designer) introduces the design and its purpose and context.
- If appropriate, the presenter gives a step-by-step overview of workflow or broader theme of the design being presented.
- The presenter gives a guided tour of the design in action by narrating the action and doing the required manipulation of wireframes and storyboards.
- Questions, comments, and discussion are invited from the entire group.
- The note taker records design issues that arise.

To be realistic and engaging, UX evaluators can explore early UX designs through the lens of design scenarios. The leader walks the group through key workflow patterns that the system is intended to support.

As the team follows the scenarios, looking systematically at parts of the design and discussing the merits and potential problems, the leader tells stories about users and usage, user intentions and actions, and expected outcomes. The leader explains what the user will be doing, what the user may be thinking, and how the task fits in the work practice, workflow, and context. Other team members may comment on how well it works for users.

Sometimes in agile practice, the UX team is tasked with updating or adding a particular feature to an existing system. Even though most of the system is in a state of high maturity, a design walk-through is a good way to get fast feedback on a design solution for this new feature and how it fits in with the existing context.

23.3.3.1 The value of click-through prototypes

A click-through prototype is (usually) a wireframe prototype that shows state progressions due to interaction inputs by users. By clicking on a link within the wireframe screen design, one can demonstrate state changes or navigation

of workflow somewhat realistically by changing the display from one wireframe screen to another.

Although a design walk-through is usually employed before design prototypes exist, even the earliest and lowest fidelity of wireframe prototypes can be made click-through by adding these links. A high-level click-through prototype could be used to demonstrate sequencing and workflow without addressing the details of design. Even at a very low level, the added impression of realism of click-through operation can be striking, giving the feeling that, for the first time, a real system is being observed.

23.4 EXPERT UX INSPECTION

23.4.1 What Is UX Inspection?

UX inspection is an analytic formative evaluation method, most appropriate when design maturity is medium to relatively high. A UX inspection involves a UX evaluation expert examining and trying out the design instead of having participants exercise it while the evaluator observes. Inspections are used to find UX design problems to refine an existing design.

UX inspection is one of the most important activities of UX evaluation and is part of the daily business of UX practitioners. Whereas a design walk-through usually entails a presentation of UX designs to an audience of stakeholders, UX inspections do not involve presentations or audiences. Inspection reports are generally not shared with anyone outside the UX team.

Evaluators are UX experts who ask themselves questions about what would cause users problems, sometimes in the presence of a set of abstracted design guidelines, and give opinions predicting UX problems.

Often inspectors simulate performance of key user tasks as a setting to guide parts of an inspection in which to judge user actions and system responses. UX inspections can be performed on designs at almost any level of maturity, from wireframe prototypes with live state transitions and screen designs for each wireframe to fully functional products or system prototypes.

A UX inspection is something like a vehicle safety inspection by an expert automotive technician. Driven by a written or mental list of mechanisms, features, and devices to check, the technician thoroughly inspects the vehicle, touching on all areas that could have problems, resulting in a list of problems to be addressed and fixed.

23.4.2 Broad Range of UX Inspection Applicability

UX inspections are especially useful for evaluation of design aspects, such as the following:

- User workflow and navigation through key user tasks
- Detailed interaction designs, including task flow, design metaphors, information objects, and design patterns
- Exception scenarios, error avoidance, and recovery mechanisms
- Compliance with UX standards or style guides
- Interaction design to ensure user journeys through the envisioned solution space are supported well
- Visual, auditory, motion (animation, etc.) details, typically involving emotional impact concerns
- Detailed evaluation of interaction, workflow, and navigation
- Iteratively evaluating task-level designs in the late funnel
- Use of domain-appropriate language
- Compliance with accessibility and internationalization policies of the organization

23.4.3 What Kind of Expertise Is Used?

Not surprisingly, Nielsen (1992) found that UX experts (UX practitioners or consultants) make the best inspection evaluators, which is why this kind of evaluation method is also sometimes called expert evaluation or expert inspection.

To perform an effective inspection, it helps if the inspector has a good knowledge of the system domain. If you are fortunate enough to have all of this expertise in one person, you have a dual expert. Otherwise, the system domain expertise can be filled in by teaming up with a system domain expert.

23.4.4 How Many Inspectors or Inspections Are Needed?

In empirical UX testing, you can improve evaluation effectiveness by adding more participants until you get diminishing returns. Similarly, in UX inspection, to improve effectiveness, you can add more inspectors or more inspections. A team approach is beneficial because low individual problem detection rates can limit the breadth of problems found by one person.

Research and experience have shown that different evaluators (even experts) find different problems, and this diversity of skills is valuable because the union of problems found over a group of inspectors is usually much larger than the set of problems found by any individual. The tradeoff, however, is cost and time. Nielsen and Landauer (1993) found that, under

some conditions, a small set of experts, in the range of three to five, is optimal before diminishing returns. If the entire UX team works together on an inspection, rather than separately, then individual differences in detection rates would not be a drawback.

In our own practice, we have found the sweet spot to be a team of two UX inspectors working together. As a practical matter, as with almost any kind of evaluation, some is better than none. Sometimes in an agile environment, one inspection by one person is all you get.

Sometimes, the same design is critiqued again after the design team makes adjustments based on feedback from the previous round, as is done in the RITE empirical evaluation method (see [22.7.3](#)).

Two specific UX inspection methods are described here in the next two sections.

23.5 HEURISTIC EVALUATION, A UX INSPECTION METHOD

Heuristic UX evaluation is one particular analytic evaluation method based on expert UX inspection, in which the evaluator compares aspects of the design against a set of heuristics, general high-level UX design guidelines.

23.5.1 Introduction

As Nielsen (Nielsen, 1992; Nielsen & Molich, 1990) stated, the heuristic evaluation (HE) method has the advantages of being inexpensive, intuitive, adaptable, and easy to motivate practitioners to do, and it is effective for use early in the UX process. It is no surprise that, of all the inspection methods, the HE method is the best known and the most popular. For the less experienced UX practitioner, performing a UX inspection based on a checklist of questions to answer is more likely to succeed than an open-ended quest to find UX problems.

23.5.2 The Heuristics

Following publication of the original heuristics (Molich & Nielsen, 1990; Nielsen & Molich, 1990), Nielsen (1994) enhanced the heuristics with a study based on factor analysis of a large number of real usability problems. The resulting refined heuristics are as follows (used courtesy of the Association for Computing Machinery):

1. **Visibility of system status:** The system should always keep users informed about what is going on through appropriate feedback within reasonable time.

2. Match between system and the real world: The system should speak the users' language, with words, phrases, and concepts familiar to the user rather than system-oriented terms. Follow real-world conventions, making information appear in a natural and logical order.
3. User control and freedom: Users often choose system functions by mistake and will need a clearly marked emergency exit to leave the unwanted state without having to go through an extended dialogue. Support undo and redo.
4. Consistency and standards: Users should not have to wonder whether different words, situations, or actions mean the same thing. Follow platform conventions.
5. Error prevention: Even better than good error messages is a careful design that prevents a problem from occurring in the first place. Either eliminate error-prone conditions or check for them and present users with a confirmation option before they commit to the action.
6. Recognition rather than recall: Minimize the user's memory load by making objects, actions, and options visible. The user should not have to remember information from one part of the dialogue to another. Instructions for use of the system should be visible or easily retrievable whenever appropriate.
7. Flexibility and efficiency of use: Accelerators, unseen by the novice user, may often speed up the interaction for the expert user such that the system can cater to both inexperienced and experienced users. Allow users to tailor frequent actions.
8. Aesthetic and minimalist design: Dialogues should not contain information that is irrelevant or rarely needed. Every extra unit of information in a dialogue competes with the relevant units of information and diminishes their relative visibility.
9. Help users recognize, diagnose, and recover from errors: Error messages should be expressed in plain language (no codes), indicate the problem precisely, and suggest a solution constructively.
10. Help and documentation: Even though it is better if the system can be used without documentation, it may be necessary to provide help and documentation. Any such information should be easy to search, focused on the user's task, list concrete steps to be carried out, and not be too large.

23.5.3 The Procedure

Despite the large number of variations in practice, we endeavor to describe what roughly represents the plain or standard version. These inspection sessions can take a couple of hours for small systems to several days for larger systems. Following is the procedure:

- The project team or manager selects a set of evaluators, typically three to five.
- The team selects a small tractable set, about 10, of heuristics, generalized and simplified design guidelines in the form of inspection questions (e.g., "Does the interaction design use the natural language that is familiar to the target user?").

- The set of heuristics given in the previous section are a good start.
- Each inspector individually browses through each part of the interaction design, asking the heuristic questions about that part and, for each heuristic question:
 - Assesses the compliance of each part of the design
 - Notes places where a heuristic is violated as candidate usability problems
 - Notes places where heuristics are supported (things done well)
 - Identifies the context of each instance noted previously, usually by capturing an image of the screen or part of the screen where the problem or good design feature occurs
- All the inspectors get together and, as a team, they:
 - Merge their problem lists
 - Select the most important ones to fix
 - Brainstorm suggested solutions
 - Decide on recommendations for the designers based on the most frequently visited screens, screens with the most usability problems, guidelines violated most often, and resources available to make changes
- Issue a group report

A heuristic evaluation report should:

- Begin with an overview of the system being evaluated
- Give an overview explanation of inspection process
- List the inspection questions based on heuristics used
- Report on potential usability problems revealed by the inspection:
 - By heuristic: For each heuristic, give examples of design violations and of ways the design supports the heuristic.
 - By part of the design: For each part, give specific examples of heuristics violated and/or supported. We recommend this approach because almost always a single part of the design violates multiple heuristics, and this structure works to fix all of them at once. Also, it is easier to walk-through findings during reporting by going through the flow and calling out everything that is broken at each step.
- Include as many illustrative screen images or other visual examples as possible.

The team then puts forward the recommendations they agreed on for design modifications, using language that will motivate others to want to make these changes. They highlight a realistic list of the average top three suggestions for modifications, and they prioritize suggestions to give the greatest improvement in usability for the least cost (perhaps using the cost-importance analysis of [Section 24.4](#)).

23.5.4 Documenting UX Problems

We have found it best to keep HE problem documentation simple. Long forms with lots of fields can capture more information, but they tend to be tedious for UX professionals who have to deal with large numbers of problems. Fig. 23.2 shows a simple HE data capture form that we have adapted, with permission, from one developed by Brad Myers.¹

Heuristic Evaluation Report	
Date: mm/dd/yyyy	
Prepared By:	Name: Signature:
Problem number: 1	
Name of heuristic violated or supported: Consistency	
Prototype screen, page, location of problem:	
Selected prototype screen image:	
	
Reason for reporting as negative or positive: Inconsistent placement of “Add to Cart” buttons: The “Add to Cart” button is below the item in CDW but above in CDW-G.	
Scope of problem: Every product page	
Severity of problem (high/medium/low): Low—minor, cosmetic problem	
Justification for severity rating: Unlikely that users will have trouble with finding or recognizing the button	
Suggestions to fix: Move the button on one of the sites to be in the same place as on the other site.	
Possible tradeoffs (why fix might not work): This may result in an inconsistency with something else, but unknown what that might be.	

Fig. 23.2
Simple heuristic evaluation
reporting form (courtesy of
Brad Myers.).

Be specific and insightful; include subtlety and depth. Saying “The system does not have good color choices because it does not use color” is trivial and not helpful. If you evaluated a prototype, then reporting that some functions are not implemented is obvious and not useful.

23.5.5 Variations Abound

The one constant about the HE method and most other analytic methods is the variation with which they are used in practice. These methods are adapted and customized by almost every team that uses them, often in undocumented and unpublished ways.

¹Private communication

As an example of a variation that was described in the literature, participatory heuristic evaluation extends the HE method with additional heuristics to address broader issues of task and workflow, beyond just the design of user interface artifacts to “consider how the system can contribute to human goals and human experience” (Muller et al., 1998, p. 16). The definitive difference in participatory HE is the addition of users and work domain experts to the inspection team.

Sears (1997) extended the HE method with what he calls heuristic walk-throughs. Several lists are prepared and given to each practitioner doing the inspection: user tasks, inspection heuristics, and thought-focusing questions. Each inspector performs two inspections, one using the tasks as a guide and supported by the thought-focusing questions. The second inspection is the more traditional kind, using the heuristics. Their studies showed that heuristic walk-throughs resulted in finding more problems than cognitive walk-throughs (Wharton et al., 1994) and fewer false positives than heuristic evaluations. A cognitive walk-through is a task-specific usability inspection method used to find usability problems, especially relating to ease of use for new users.

Cockton, Lavery, and Woolrych (2003) developed an extended problem-reporting format that improves heuristic inspection methods by finding and eliminating many of the false positives typical of the usability inspection approach. Their discovery and analysis resource model allows analysts to bring distinct discovery and analysis resources together to isolate and analyze false negatives and false positives.

23.5.6 Limitations

Although a helpful guide for inexperienced practitioners, we find that heuristics often get in the way of the experts. To be fair, the heuristic method was intended as a kind of scaffolding to help novice practitioners do usability inspections, so it should not be compared with expert usability inspection methods. In addition, the heuristics are very general and defined at a high level, limited in their descriptive power by a lack in depth of coverage and specificity.

It was perhaps self-confirming when we read that others found the heuristics to be similarly unhelpful (Cockton, Lavery, & Woolrych, 2003; Cockton & Woolrych, 2001). In their studies, Cockton and Woolrych (2003) found that it is experts who best find problems with inspection based on broad knowledge and experience, not experts using heuristics. Cockton and Woolrych (2002, p. 15) also claim that the “inspection methods do not encourage analysts to take a rich or comprehensive view of interaction.” Although this may be true for heuristic methods, it does not have to be true for all inspection methods.

A major drawback with any inspection method, including the HE method, is the danger that novice practitioners will get too comfortable with it and think the heuristics are enough for any evaluation situation. There are few indications in its usage that let the novice practitioner know when it is not working well, and when a different method should be tried.

Also, like all UX inspection methods, the HE method can generate numerous situations in which inspectors identified problems that turned out to be not real problems or not very important UX problems. Finally, like most other analytic UX evaluation methods, the HE method is not always effective in finding usability problems below the surface—problems about sequencing and workflow.

23.6 OUR PRACTICAL APPROACH TO UX INSPECTION

We have synthesized existing UX inspection methods into a relatively simple and straightforward method that, unlike the heuristic method, is definitely for UX experts and not for novices. Sometimes, we have novices sit in and observe the process as a kind of apprentice training, but they do not perform these inspections on their own.

23.6.1 Let Insight and Experience Be Your Guide

We do not directly or explicitly use a list of heuristics to drive this kind of UX inspection. In our own industry and consulting experience, we have not found specific heuristics as useful as we would like. We drive our inspection process with usage in context by focusing our UX knowledge and experience on tasks and work activities. We bring our expert knowledge of UX design guidelines (see [Chapters 27 and 29](#)) to bear on issues that are real problems and to understand the underlying nature of the problems and potential solutions.

Using this approach, and our UX intuition honed over the years, we can see, and even anticipate, UX problems, many of which may not have been revealed under a purely heuristic spotlight.

23.6.2 Use a Co-discovery Approach

Expert UX professionals as inspectors are in the role of UX detectives. To aid the detective work, it can help to team up two practitioners in co-discovery (O'Malley, Draper, & Riley, 1984), who work together as mutual sounding boards in a give-and-take interplay, to potentiate each other's efforts to keep the process flowing, to promote a constant flow of think-aloud comments from the inspectors, and to maintain a barrage of problem notes flying.

It can be useful to add other roles to this team (e.g., customers, users, designers, others familiar with the overall system), who can help make up for any lack of system knowledge on your part, especially if you have not been with the team during the entire project. Work domain experts can reinforce your user-surrogate role and bring in more subject-matter expertise (Muller et al., 1998).

To follow this to a logical conclusion, it can also be beneficial to use the whole UX team as inspectors, although a large number of inspectors cannot have the same intimate interplay as two inspectors.

23.6.3 Explore Systematically With a Rich and Comprehensive Usage-Oriented View

As an inspector, you should not just look for individual problems associated with tasks or functions. Use all your experience and knowledge to see the big picture. Keep an expert eye on the high-level view of workflow, the overall integration of functionality, and emotional impact factors that go beyond usability.

23.6.4 Inspection Can Be Driven by Tasks and by the Design Itself

Representative user tasks help us put ourselves in the users' shoes. By exploring the tasks ourselves and taking our own think-aloud data, we can imagine what real users may encounter in their usage. A hierarchical task inventory (see [Section 9.2](#)) is helpful in attaining a good understanding of the task structure and ensuring broad coverage of the range of tasks. Usage scenarios and design scenarios (see [Section 9.3.1](#)) are fruitful ways to focus on key user work roles and key user tasks that must be supported in the design.

Driving the inspection with the interaction design itself means trying all possible actions on all the user interface artifacts and trying out all user interface objects, such as buttons, icons, and menus. It also means being opportunistic in following leads and hunches triggered by parts of the design.

23.7 EMOTIONAL IMPACT INSPECTION

In the past, inspections for evaluating UX designs have been almost exclusively usability inspections at the interaction level. This kind of evaluation can also be very effective at addressing issues of emotional impact. The process is essentially the same, but you need to look beyond a task view to the overall usage experience.

Among the emotional impact questions to have in mind in a UX inspection are:

- Is usage fun?
- Is the visual design attractive (e.g., colors, shapes, layout) and creative?
- Will the design delight the user visually, aurally, or tactilely?
- If the target is a product:
 - Is the packaging and product presentation aesthetic?
 - Is the out-of-the-box experience exciting?
 - Does the product feel robust and good to hold?
 - Can the product add to the user's self-esteem?
 - Does the product embody environmental and sustainable practices?
 - Does the product convey the branding of the organization?
 - Does the brand stand for progressive, social, and civic values?
 - Are there opportunities to improve emotional impact in any of the aforementioned areas?

You can also use focus groups (see [Section 6.6.1](#)) and/or co-discovery to evaluate the emotional layer. Most of the questions in a questionnaire for assessing emotional impact are also applicable as inspection questions here. As an example, using attributes from AttrakDiff (Hassenzahl, Burmester, & Koller, 2003):

- Is the system or product interesting?
- Is it exciting?
- Is it innovative?
- Is it engaging?
- Is it motivating?
- Is it desirable?

Exercise 23.1 Analytic Evaluation (UX Inspection) of Your Product or System

Goal: To practice doing a UX inspection (see [Section 5.6.1.2](#)).

Activities: It is strongly preferred that you use the wireframe prototype deck you built in [Exercise 18.1](#). If your prototype is not suitable for an effective exercise in UX inspection, then select an application or appropriate website as the target of your inspection.

Perform a team-based UX inspection as described in this chapter.

Deliverables: Your list of inspection questions, if you used one. A list of UX problems identified by your UX inspection.

Schedule: About 90 minutes.

23.8 SEGUE

In this chapter and the previous one, we covered UX evaluation data collection using both empirical and analytical methods. In the next chapter, we move on to the discussion about analyzing these data.